FE101 · Introduction to TypeScript & Modern JavaScript
Clock Hours: 45
Quarter Credit Hours: 4
Prerequisites: PY104
Instructors
Angela Cook
B.S. in Computer Software Development, Western Governors University.
Bradley Smith
M.S. in Statistics, University of Nebraska–Lincoln – 10 yrs in software engineering. Also holds Ph.D. in Psychological Studies in Education, University of Nebraska–Lincoln.
chris Esposito
Ph.D. in Computer Science, New Mexico State University – 38 yrs in software engineering.
Clifford Osburn
A.A.S. in CIS Software Systems & Networking, Amarillo College – 25 yrs in software engineering.
David Gross
M.S. in Computer Engineering, University of Central Florida – 20 yrs in software engineering.
Dr. David Votaw
DCS in Computer Science (Information Assurance), Colorado Technical University – 16 yrs in software engineering.
Dr. Xuan Dung (James) Nguyen
9 yrs in software engineering: Python/ML through a Virginia Tech PhD; grad TA mentoring 50+ undergrads (2020–25); VT grad cert. in Data Analytics. No degree in field; holds Ph.D. in Chemical Engineering, Virginia Tech.
ed one
Frank Chambergo
B.S. in Information Science, University of Maryland – 7 yrs in software engineering.
Gregory Keeney
B.S. in Computer Science, Northwest Missouri State University – 9 yrs in software engineering.
Ishay Gavriel
John Berta
12 yrs in software engineering: AP Computer Science instructor since 2014 (Hazleton Area SD); Python engineer at BrickSeek (2018–24); UPenn cert. in Applied Computational Thinking. No degree in field; holds M.A. in Classroom Technology, Wilkes University.
Kris Jamsa
Ph.D. in Computer Science, Arizona State University – 18 yrs in software engineering.
Neil Kalinowski

M.S. in Computer Science, University of Texas at San Antonio – 26 yrs in software engineering.
Shourav Mandal
M.S. in Computer Science (AI & Machine Learning), Colorado State University – 5 yrs in software engineering.
Vlad Blanton
10 yrs in software engineering: full-stack engineer at CasaLP (2025–) and freelance React/Node (2020–25); CS/Math instructor (2018–20). No degree in field; holds B.A., Reed College.
Have an academic question? Submit an Academic Assistance Request, and Student Services will make sure it reaches the right course instructor or member of the academic team.

Course Description
Introduction to TypeScript & Modern JavaScript equips students with practical knowledge and applied skills. Students will explore topics such as The History and Role of JavaScript, JavaScript Engines and Runtimes, Introducing TypeScript, Setting Up Your Development Environment, Statements, Expressions, and Blocks.

Course Objectives
JavaScript’s role in modern development
Core JavaScript syntax and data types
Objects, arrays, and functions
Asynchronous code with promises and async/await
Adding static typing with TypeScript
Course Outline
Unit Topics & Assignments
Unit 1 – From Python to JavaScript Topics:
• JavaScript across browser and server environments; console.log and semicolons
• Variable declarations with const, let, and camelCase
• Primitive types, typeof, and JavaScript coercion versus Python
• Conditional blocks and template-literal interpolation
• Strict equality, nullish and falsy values, and array basics

Assessment:
Review (quiz)
Unit 2 – JavaScript data and logic Topics:
• JavaScript objects, access notation, and Python dictionary comparisons
• while, for, and for...of loops with early exits
• Destructuring, spread, shallow copies, mutability, and references

• Function parameters and single or multiple return values
• First-class functions, callbacks, anonymous functions, and arrow syntax

Assessment:
Review (quiz)
Unit 3 – Going async Topics:
• Array transformation with forEach, map, and filter
• Synchronous and asynchronous execution, event loops, and setTimeout
• fetch(), Promises, and pending, fulfilled, or rejected states
• Promise chains, error handling, JSON parsing, and execution order
• async/await, dependent requests, and clean error termination

Assessment:
Review (quiz)
Unit 4 – A taste of TypeScript Topics:
• Static typing, JavaScript, and TypeScript analysis
• Variable type annotations, inference, and assignment safety
• Typed function parameters and argument errors
• Function return types, void, and input-output documentation

Assessment:
Final course review (quiz)
Review format
Reviews in this course consist of a MIX of open-ended and multiple-choice questions, focused on practical understanding of core concepts: applying key ideas, interpreting code, identifying errors, and reasoning about system behavior.

NOTE: The outline is subject to change at the institution’s discretion

Text & Reference Materials
Proprietary instructional content
TypeScript Docs: https://www.typescriptlang.org/docs/
Mozilla Developer Network (MDN): https://developer.mozilla.org/
Teaching Strategies
Individualized tutorials and demonstrations.
LMS Assignments.
Hands-on practice labs.
Projects.
Interactive discussions and case studies.
Grading Policies

Students will be evaluated in the following manner:

Quizzes (Reviews): 30%.
Online Assignments (Discussions): 10%.
Tests/Exams (Final Review): 60%.
Bonus: Students who submit and pass all reviews on time are eligible to earn up to 10 bonus points.

The grading percentages noted may change at the discretion of the institution.

To satisfactorily complete this course, a student must achieve a minimum of 60 percent (a grade of D). In order to graduate, a student must attain a minimum cumulative GPA of 2.0 and complete all courses with a passing grade.

Grading Scale
Final grades are based on the final grading system found in the catalog.

Grade	Numeric Grades	Grade Points
A	100 – 90	4.0
B	89 – 80	3.0
C	79 – 70	2.0
D	69 – 60	1.0
F	59 & below	0.0

Reviews
All reviews must be completed by the end of the term for course completion. At term end, grades finalize and reviews lock – the only exception is the final review retake, which is open for one week after the start of the following term.

Reviews
A formative assessment that helps students reinforce each unit’s content.

Reviews are scored, but only pass/fail affects your grade: a result of D or higher counts as a Pass and earns full points for that review; a higher passing grade doesn’t add extra course-grade points.
A result below D counts as a Fail and earns no points. A failed attempt does not complete the lesson, and you may try again.
Reviews can be retaken as long as the course remains open for learning.
Reviews are graded automatically, and results are available immediately.

Your first passing attempt completes the lesson and allows you to continue.
Discussions
A challenge, question, or exercise related to the content covered in one or more academic units.

Responses to discussions can be submitted at any time until the end of the term. You can find the term's end date in your Learning Journey timeline.
Whenever a discussion response is submitted, the final course grade is automatically recalculated and updated based on all grade components.
Final Reviews
The final course grade will be published on the later of the following dates: eight days after the course ends, or the date the exam is graded.
Students may take a retake exam; the higher of the two grades is the final exam grade. The retake window is open for one week after the start of the following term.
If a student takes the retake exam attempt, the final course grade will be updated after that exam is graded.
Instructor Availability
This course requires significant individual effort. If you experience any challenges, contact Student Services as soon as possible for academic support or to connect with your course instructor. Seeking help early can make it easier to stay on track and succeed in the course.

Student Responsibilities
This course is part of Maestro College’s asynchronous distance education program, a non-attendance-taking, academic-event-based program. Coursework is completed through the Maestro Learning Platform in an asynchronous online format, allowing students to access materials at any time throughout the week. There are no scheduled class meetings; however, students are expected to complete assignments at least weekly, meet deadlines, and participate regularly in online discussions and activities.

Attendance & Enrollment
Requirements for attendance in asynchronous distance education programs:

Maestro College Asynchronous Distance Education programs are non-attendance taking, academic event based programs. Although these programs are non-attendance taking, the institution is required to document students’ participation

in academic related activity and to terminate the enrollment of any student for which this activity cannot be documented at the following points:

at the end of the first calendar week of the academic term;
at the end of the first four calendar weeks of the academic term;
at the midpoint of each academic term; and
at the end of each academic term.
Example: For a program which is run in 10-week terms a student must complete a qualifying academic activity in weeks 1, 4, 5, and 10 to remain active. Failure to complete a qualifying academic activity during any of these weeks will result in the student being dropped from the program.

Samples of qualifying academic events include:

Attending a synchronous class, lecture, recitation, or field or laboratory activity, physically or online, where there is an opportunity for interaction between the instructor and students
Submitting an academic assignment
Taking an assessment or an exam
Participating in an interactive tutorial, webinar, or other interactive computer-assisted instruction
Participating in a study group, group project, or an online discussion that is assigned by the institution
Interacting with an instructor about academic matters
The following do NOT qualify as academic events:

Logging into an online class or tutorial without any further participation; or
Participating in academic counseling or advisement.
Academic Support
Students enrolled in asynchronous online programs receive academic support through structured, asynchronous support channels aligned with the online instructional model. To request academic assistance, students must submit an academic support ticket through the Maestro Platform, which is administered by Student Services. Academic questions submitted through this system are routed to qualified instructional staff, who provide clarification, guidance, or additional explanation related to course content.

Academic Integrity
You are encouraged to discuss course concepts and assignments with classmates to deepen your understanding, but all

submitted work must be your own. Do not copy another person’s work or submit or share solutions that are not your own.

If an academic integrity violation is confirmed, the College may impose sanctions, as determined by the appropriate academic authority and as outlined in the school catalog.

AI Usage
Students may use AI tools for non-graded academic support, such as:

Reviewing material
Asking practice questions
Practicing skills
Students may not:

Use AI to generate graded assignments unless explicitly permitted
Present AI-generated answers as original work without citation
AI misuse may be treated as academic dishonesty.

NOTE: This syllabus is subject to change at the institution’s discretion